



Computer Networks — Switching Techniques

GATE-Oriented Notes

1. Switching Network — Basic Idea

A communication network generally consists of **end systems** connected through **intermediate switching nodes**.

If the source and destination are directly connected, communication can take place directly.

However, when they are far apart, data must pass through multiple intermediate nodes.

Source → Switch → Switch → Switch → Destination

S → N1 → N2 → N3 → D

Functions of Switching Nodes

- Provide/establish a path between source and destination.
- Forward data from one node to the next.
- Determine the appropriate route when required.
- Generally, switching nodes are concerned with **switching/forwarding**, not with the actual content of the data.

GATE Definition

A **switching network** consists of end systems interconnected through intermediate switching nodes, through which data is routed from the source to the destination.

Important Points

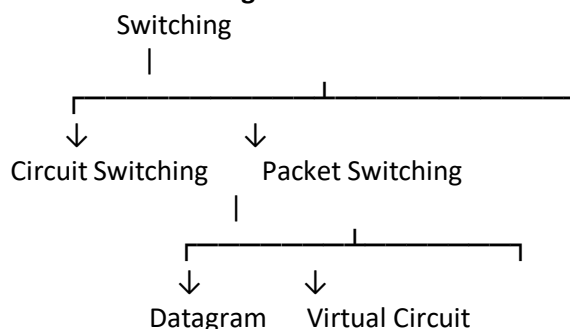
- A network may have **multiple possible paths**.
- Redundant paths can improve reliability.
- Failure of a node or link may require an alternative path.

2. Major Switching Technologies ★ ★ ★

There are two predominant switching technologies:

1. Circuit Switching

2. Packet Switching



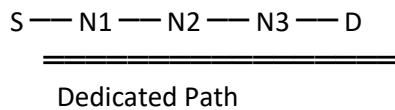
3. Circuit Switching ★ ★ ★

Definition

In **circuit switching**, a **dedicated communication path** is established between the source and destination before data transfer begins.

Once established, the required resources along the path are reserved for that communication.

Example: Traditional telephone networks.



4. Three Phases of Circuit Switching ★ ★ ★

Circuit switching consists of three phases:

1. Connection Establishment / Path Establishment

- A path is established between source and destination.
- Required resources are allocated/reserved.
- The routing path is determined.

2. Data Transfer

- Data is transmitted through the established dedicated path.
- Communication can continue as long as the connection exists.

3. Connection Teardown / Disconnection

- The connection is terminated after communication.
- Reserved resources are released.

Remember

ESTABLISH → TRANSFER → TEARDOWN

↓ ↓ ↓
Reserve Data Release
resources transfer resources

5. Requirements of Circuit Switching

For establishing a circuit, the network requires:

1. Switching Capacity

The switching network must have sufficient switching resources to establish the required connection.

2. Channel Capacity

The selected communication channels must have sufficient capacity to carry the data.

3. Routing Capability

The network must determine a suitable path from source to destination.

The path may be:

- Predefined, or
- Determined using a routing algorithm/mechanism.

6. Blocking and Non-Blocking Architecture ★

Circuit switching has limited resources.

Suppose a network can support **10 simultaneous connections**.

If all 10 connections are already occupied and another connection request arrives:

Blocking

The new connection cannot be established because the required resources are unavailable.

Non-Blocking

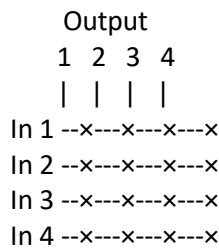
The network architecture is designed so that a requested connection can be established whenever the required resources are available through the switching structure.

GATE Focus

Blocking is related to the **availability of a suitable path/resources** for establishing a connection.

7. Circuit Switching — Space Division Switching

In **Space Division Switching**, physical switching paths are established using switching elements/crosspoints.



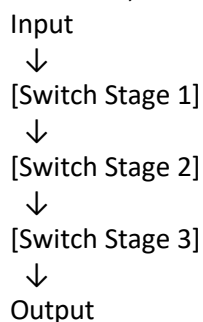
A **controller** determines which input should be connected to which output.

Key Idea ★

Space Division Switching → Physical paths/crosspoints are used for switching.

8. Multi-Stage Space Division Switching

A single-stage switching matrix can require a large number of crosspoints for a large network. Therefore, multiple smaller switching stages can be used.



Advantages

- Can reduce the number of switching elements/crosspoints.
- Provides a more efficient structure for large switching networks.

Important

Multi-stage switching must be analyzed for:

Blocking vs Non-Blocking behavior

A connection may fail if a suitable path through the intermediate stages is unavailable.

9. Time Division Multiplexing (TDM) Switching

In **Time Division Switching**, switching is performed using **time slots**.

Different data streams are assigned different time slots.



The order of time slots can be changed depending on the required connection.

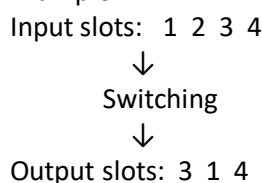
Key Idea ★

Time Division Switching → Switching is based on time slots.

10. Time Slot Interchange (TSI)

Time Slot Interchange allows incoming time slots to be rearranged/selectively assigned to outgoing time slots.

Example:



A control mechanism determines which incoming time slot should appear at a particular outgoing

position.

11. Circuit Switching — Properties and Issues



1. Dedicated Connection

Once the connection is established, the communication uses the reserved resources along the established path.

Therefore, communication becomes relatively **transparent** after setup.

2. Mainly Developed for Voice Traffic

Circuit switching was primarily developed for **voice communication**, especially traditional telephone networks.

3. Bandwidth Wastage ⚠️

The channel remains reserved for the duration of the connection.

Even when no data is being transmitted:

Reserved Bandwidth



Still Occupied



Cannot be used by another connection

Therefore, bandwidth may be wasted.

4. Connection Setup Delay

A connection must first be established before data transfer can begin.

Connection Establishment



Data Transfer

Thus, circuit switching introduces **setup delay**.

5. Fixed Data Rate ★

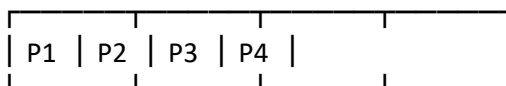
The established circuit operates at a fixed data rate, so the communicating endpoints need to operate at compatible data rates.

12. Packet Switching ★ ★ ★

Definition

In **packet switching**, a long message is divided into a number of smaller units called **packets**.

Original Message



Network



Destination

Each packet contains:

- **Data**
- **Control information**

Control information is mainly used for **addressing and routing**.

13. Store-and-Forward Mechanism ★ ★ ★

In packet switching, an intermediate node generally performs:

Receive → Store/Buffer → Forward

For example:

$S \rightarrow N1 \rightarrow N2 \rightarrow N3 \rightarrow D$

At N1:

Receive Packet



Store/Buffer



Forward to N2

Thus, packets are temporarily stored at intermediate nodes before being forwarded.

14. Advantages of Packet Switching ★ ★ ★

1. Better Line Efficiency

A physical link can be shared by packets belonging to different transmissions.

There is no permanent dedicated circuit.

2. Dynamic Resource Sharing

Network resources are allocated according to demand.

3. Data Rate Conversion

Different stations/links may operate at different data rates.

Intermediate nodes can buffer packets and help handle these rate differences.

4. Handling Busy Networks

Packets can be accepted and temporarily queued even when the network is busy.

However, congestion can increase delay.

5. Packet Prioritization

Packets can be assigned different priorities so that higher-priority packets can be handled first.

15. Packet Switching Techniques ★ ★ ★

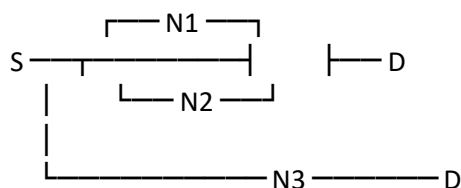
Two major packet-switching techniques are:

1. **Datagram**
2. **Virtual Circuit**

16. Datagram Packet Switching ★ ★ ★

In the **datagram approach**, every packet is treated independently.

Each packet can take an appropriate route through the network.



For example:

P1 → Route 1

P2 → Route 2

P3 → Route 3

P4 → Route 1

Characteristics

- No connection setup is required.
- Each packet is independently routed.
- Packets may follow different routes.
- Packets may arrive **out of order**.
- Packets may be delayed.
- Packets may be lost.
- The receiver may need to **reorder packets**.

- Missing packets may require retransmission mechanisms.

★ GATE Trap

Datagram does NOT guarantee in-order delivery.

Example:

Sent: P1 → P2 → P3 → P4

Received: P2 → P1 → P4 → P3

17. Virtual Circuit Packet Switching ★ ★ ★

In **Virtual Circuit (VC)** packet switching, a logical route is established before packet transfer.

Typical sequence:

Connection Request



Call Accept



Virtual Circuit Established



Packet Transfer



Circuit Release

Important

A virtual circuit is a **logical connection**.

It does **not necessarily mean a physically dedicated path**.

Multiple virtual circuits may share the same physical resources.

18. VCI — Virtual Circuit Identifier ★ ★ ★

Packets in a virtual circuit network carry a **VCI (Virtual Circuit Identifier)**.

Intermediate switches use the VCI and port information to determine where the packet should be forwarded.

Example:

Incoming:

Port 1 + VCI 77



Switch Table



Outgoing:

Port 3 + VCI 22

The VCI can be changed from one link/node to another.

Important GATE Point ★ ★ ★

VCI has local significance.

Therefore, the same VCI value can theoretically be reused in different local portions of the network.

19. Datagram vs Virtual Circuit ★ ★ ★ ★ ★

Feature	Datagram	Virtual Circuit
Connection setup	✗ Not required	☑ Required
Route	Independently determined for packets	Logical route established
Routing decision	For each packet	Mainly during setup
Packet order	May be out of order	Sequencing can be supported
Packet forwarding	More routing work per packet	Faster after setup
Failure handling	Can potentially use another route	May require VC re-establishment

Identifier	Destination/address information	VCI
Setup overhead	Low	Higher
Routing flexibility	High	Lower after VC establishment
Suitable for	Intermittent/fewer packets	Continuous/repeated packet flow

20. Circuit Switching vs Packet Switching



Feature	Circuit Switching	Packet Switching
Path	Dedicated	Shared
Connection setup	Required	Datagram: Not required
Bandwidth	Guaranteed/reserved	Dynamically allocated
Resource utilization	May be inefficient	More efficient sharing
Data	Continuous stream	Divided into packets
Setup delay	Present	Datagram: No call setup
Data rate	Fixed	Can accommodate different rates using buffering
Congestion	Circuit capacity is not affected by other traffic	Queueing/congestion may occur
Resource usage	Reserved for entire connection	Used as needed
Typical application	Traditional voice/telephone	Computer/data communication

The lecture emphasizes that circuit switching provides reserved/guaranteed bandwidth, while packet switching dynamically shares physical channels and is primarily used for data communication.

21. GATE — Most Important Points

★ Circuit Switching

ESTABLISH → TRANSFER → TEARDOWN

Keywords:

- Dedicated path
- Resource reservation
- Setup delay
- Fixed data rate
- Guaranteed bandwidth
- Traditional voice communication

★ Packet Switching

Message → Packets → Network → Destination

Keywords:

- Small packets
- Shared resources
- Dynamic allocation
- Store-and-forward
- Buffering
- Possible congestion/delay

★ Datagram

No setup



Independent packets



Independent routing



Possible out-of-order delivery

★ Virtual Circuit

Setup



Logical route



VCI-based forwarding



Packet transfer



Release

⚠ GATE Common Traps

Trap 1

All packet-switched packets follow different paths.

✗ False.

Different paths are possible in **datagram switching**. Virtual circuits follow the established logical route.

Trap 2

Virtual circuit means a physically dedicated circuit.

✗ False.

A VC is a **logical connection**; physical resources may be shared.

Trap 3

Circuit switching has no delay.

✗ False.

It has **connection establishment/setup delay**.

Trap 4

Packet switching guarantees bandwidth.

✗ False.

Bandwidth is dynamically allocated and congestion may occur.

Trap 5

Datagram packets always arrive in sequence.

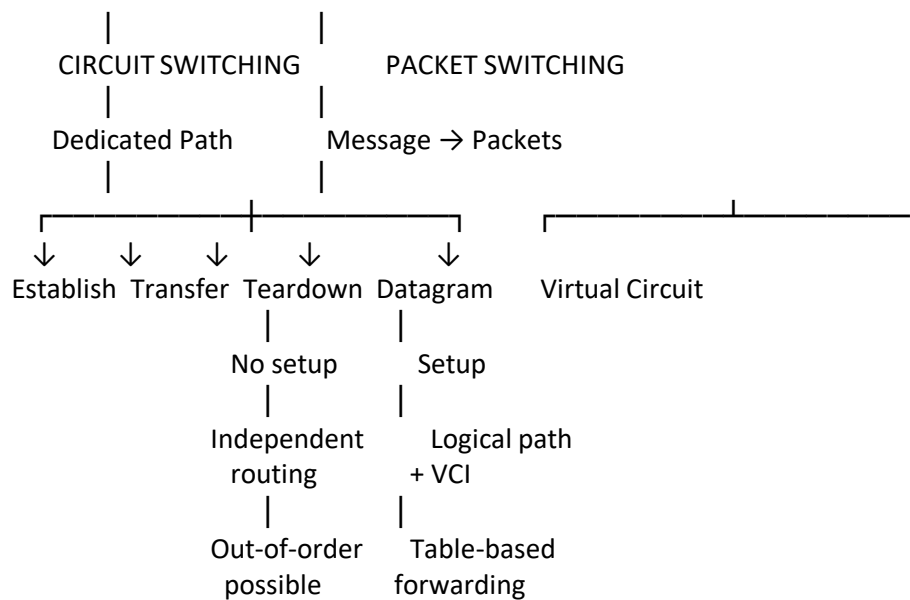
✗ False.

Packets can follow different paths and therefore may arrive out of order.

🔄 One-Minute GATE Revision

SWITCHING





🏆 Final GATE Memory Lines

Circuit Switching = Reserve resources first, then communicate.

Packet Switching = Share resources dynamically among packets.

Datagram = No setup + independent packet routing + possible out-of-order delivery.

Virtual Circuit = Setup + logical path + VCI-based forwarding.

Packet Switching uses Store-and-Forward: Receive → Store/Buffer → Forward.